UNIVERSITY OF OULU

FACULTY OF INFORMATION TECHNOLOGY AND
ELECTRICAL ENGINEERING

Big Data Processing and Applications

EXPLORATORY AND SENTIMENT ANALYSIS OF YELP REVIEWS USING APACHE SPARK

GROUP 12

ARSHMAN TARIQ
MUHAMMAD FAIZAN TANVEER

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1 Project Description

Online review platforms such as Yelp generate vast volumes of user-generated reviews that carry rich patterns about customer satisfaction, business quality, and consumer behaviour. The Yelp Open Dataset—a publicly available academic resource—spans more than 8.5 GB of structured JSON data comprising roughly seven million reviews, 150,000 businesses, and two million user profiles across metropolitan areas in the United States and Canada. Its scale, multi-table structure, and blend of textual and numerical attributes make it suitable for Big Data analytics using Apache Spark.

The motivation for this project is threefold:

1. **Business Dependency:** Businesses increasingly depend on automated review analytics to steer decisions about service quality, menu offerings, and marketing; manual inspection of millions of reviews is neither feasible nor scalable.
2. **Beyond Star Ratings:** While star ratings provide a summary signal, the review text contains richer, aspect-level cues (food quality, waiting time, staff attitude) that numeric ratings alone cannot capture—making NLP-based sentiment analysis a necessary complement.
3. **Big Data Application:** The project provides a practical opportunity to exercise the full Spark ecosystem—DataFrame API, Spark SQL, MLlib, and distributed visualisation—on a real-world dataset that exceeds single-node capacity.

The project pursues three research questions. **(RQ1)** What geographic, temporal, and categorical patterns characterise Yelp businesses and their reviews? **(RQ2)** Can ML models trained on combined textual and meta-features accurately classify review sentiment as Negative, Neutral, or Positive? **(RQ3)** Does model-derived sentiment correlate with real-world business outcomes, i.e., the likelihood of a business closing?

2 Related Work

Prior research on Yelp and related review datasets falls into three broad streams:

- Classical machine learning for classification
- Deep learning and transformer-based approaches
- Topic-modelling-driven business analytics

Classical machine learning. Sanchez [1] trained a Linear Support Vector Classifier (LinearSVC) with TF-IDF features on two million Yelp reviews, achieving 87.5% three-class accuracy. That work highlighted a persistent challenge: the Yelp corpus is heavily skewed toward positive ratings (4–5 stars), causing models to under-classify neutral and negative reviews without explicit class balancing—a finding directly relevant to our experimental design. Sisal [2] approached the same dataset from an industrial recommendation-system perspective, comparing KNN (supervised, using TF-IDF and BERT features) against DBScan and K-means clustering. KNN reached 82% accuracy while the unsupervised methods remained below 50%, demonstrating that labelled sentiment data substantially improves recommendation quality.

Deep learning and transformers. Sadikin and Fauzan [3] compared LSTM and MLP architectures on 100,000 balanced Yelp restaurant reviews, using Word2vec embeddings for feature extraction. Their LSTM reached 91% binary accuracy (positive vs. negative) against 76% for MLP, attributing the gap to the LSTM's ability to retain long-range textual context through its memory-cell mechanism. Belaroussi et al. [4] conducted the most comprehensive recent comparison, evaluating LSTM, Bi-LSTM, and four BERT-family pipelines (DistilBERT, BERT, RoBERTa, BERTweet) on a balanced three-class Yelp dataset of 107,946 reviews. Contrary to expectation, their domain-specific LSTM outperformed all transformer pipelines (77% macro accuracy); RoBERTa and BERTweet exhibited a bias toward positive predictions, while the BERT pipeline over-predicted negative sentiment. Interestingly, the neutral class proved to be the hardest for all models—indicating an inherent mixed polarity of three-star reviews.

Topic modelling and business analytics. Zhang et al. [5] applied Latent Dirichlet Allocation (LDA) to Yelp restaurant reviews, extracting five latent topics (price, time, food, service, location) and aggregating TextBlob polarity scores per topic. Their four-phase research model demonstrated that combining topic extraction with sentiment analysis yields more actionable business intelligence than star ratings alone. Guda et al. [6] proposed a multi-task joint BERT model that simultaneously predicts star ratings and an AFINN-based sentiment score, achieving 87.8% binary accuracy. Their correlation analysis established that the AFINN sentiment score and the Yelp star rating are strongly positively correlated ($r \approx 0.52$)—a benchmark our work independently validates at the business level.

Positioning of the present work. Compared with these studies, our contribution is distinguished by: (i) using the *full* multi-table Yelp dataset rather than a review-only subset; (ii) augmenting TF-IDF text features with 15 engineered meta-features drawn from user behaviour, business engagement, and temporal context; (iii) conducting both EDA and ML analysis within a unified Spark pipeline capable of processing the full 7M-row corpus; and (iv) extending beyond classification to a downstream business reputation scoring module that links model output to observable business outcomes—an application not addressed in any of the related works cited above.

3 Data Description

The Yelp Open Dataset is publicly available and accessible for educational, and academic use at `https://www.yelp.com/dataset`. The dataset comprises five JSON files of over 8.5 GB uncompressed, well above the project's 5 GB ceiling and its 100,000 data-point floor when rows and columns are counted across all tables. Table 1 summarises each table's key fields and approximate row count as read in Apache Spark.

Structural details. Star ratings are integer-valued on a 1–5 scale. The `categories` column in the Business table is a comma-separated string of tags (e.g., "Restaurants, Pizza, Italian"); overlapping tags required a custom taxonomy to avoid double-counting (see below). The `date` fields in Review and Tip are ISO-formatted strings that were cast to `TimestampType`. The Checkin table encodes all events for a business as a single comma-delimited string, requiring a `size(split(...))` aggregation to derive a visit count. The nested `attributes` and `hours` structs in the Business table were dropped in the EDA notebook due to >30% missingness and high engineering cost relative. A

Table 1: Schema summary of the five Yelp dataset tables

Table	Key Fields	Rows	Format
Review	review_id, user_id, business_id, stars, text, date, useful, funny, cool	∼6.99M	JSON
Business	business_id, name, city, state, categories, stars, review_count, is_open, latitude, longitude, attributes, hours	∼150k	JSON (nested)
User	user_id, name, review_count, average_stars, fans, useful, funny, friends, elite, cool, yelping_since	∼2.0M	JSON
Checkin	business_id, date (CSV of timestamps)	∼131k	JSON
Tip	business_id, user_id, date, text	∼909k	JSON

Power BI entity-relationship diagram of the five tables is shown in Figure 1.

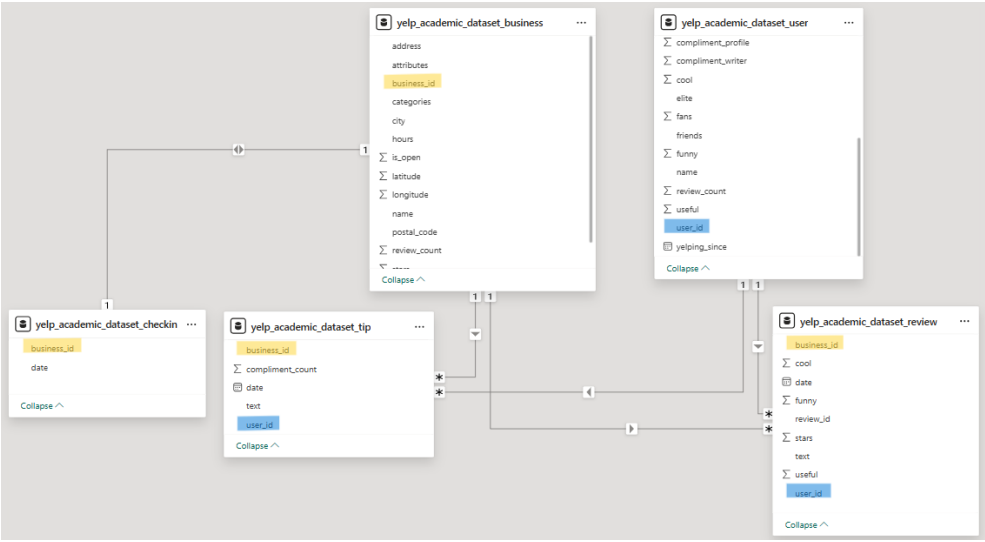


Figure 1: ERD of the five YELP tables. Primary and Foreign Keys are highlighted.

Descriptive statistics. Table 2 presents summary statistics for the most analytically relevant numerical columns, derived using PySpark's `.summary()` method.

Table 2: Descriptive statistics for key numerical features

Feature	Min	Max	Mean	Median	Std
Review stars	1	5	≈3.74	4	≈1.38
Review length (chars)	1	5000+	≈731	≈600	≈649
Review useful	0	2400+	≈1.45	0	≈9.7
Business stars	1.0	5.0	≈3.59	3.5	≈1.01
Business review count	3	12,000+	≈57	18	≈165
User review count	0	17,000+	≈25	6	≈77
User average stars	1.0	5.0	≈3.77	3.83	≈0.72
Tip compliment count	0	3900+	≈0.14	0	≈2.2

The strongly right-skewed distribution of review stars (mean 3.74, mode 5) and the long-tailed distribution of review length and review count are consistent with reports in related work [1, 6] and motivated the class-balancing in the ML pipeline.

Pre-processing. The following steps were applied, with differences between the EDA and ML pipelines noted where relevant.

Shared steps: (1) Date and Categories strings in Review, User, Tip, Business tables were parsed to `TimestampType` and `array` respectively for temporal feature extraction and time-series aggregation. (2) The `cool`, `funny`, and `useful` engagement columns in the Review table contained invalid minimum values of -1; these were replaced with 0 using a conditional expression, as no legitimate negative count is possible. (3) Rows with null or empty `categories` in the Business table (<1% of rows) were removed, since category information is essential for segmented analysis. Rows with empty `postal_code` were also dropped to preserve geographic integrity. (4) Missing street `address` values were imputed by concatenating `city` and `postal_code`, preserving geographic searchability at the cost of exact street precision.

EDA-specific: The `attributes` and `hours` nested structs were dropped from the Business table. A custom high-level category taxonomy was hand-crafted to map 200+ fine-grained Yelp tags into ten interpretable macro-categories (e.g., *Food & Dining*, *Shopping & Retail*, *Health & Medical*) using Spark's `when/contains` logic. This resolved the double-counting problem caused by Yelp listing the same venue under permutations of the same tags and mirrors the topic-grouping of Zhang et al. [5].

ML-specific: The five tables were joined into a unified analytical frame (review left-joined to business, user, checkin aggregate, business tip aggregate, and user tip aggregate). Nulls introduced by left joins were filled with 0 for numeric features and an empty array for the categories list. Three text-derived features were engineered: `review_length` (character count of the review body), `num_exclamations` (count of "!" as an emotional-intensity proxy), and `num_caps` (uppercase character count as an emphasis proxy). Two user-behaviour composites were derived: `user_activity` = `user_review_count` + `fans`, and `user_engagement` = `user_useful` + `user_funny` + `user_cool`. Year and month were extracted from the review timestamp to capture possible temporal drift in sentiment.

4 Methods and Tools

Apache Spark and platform. All analyses were executed using **Apache Spark** via the PySpark API on the *Rahti* cloud platform provided for the course. Spark's distributed in-memory execution model was essential: loading all five tables and joining them to a 6.99M-row analytical frame would exhaust the RAM of a typical laptop. Spark SQL and the DataFrame API were used interchangeably for exploratory queries (e.g., top-10 rankings via both `groupBy/orderBy` and embedded SQL views), demonstrating their equivalence while taking advantage of Catalyst optimisation. PySpark's **MLlib** library was used exclusively for the ML pipeline, keeping all transformations distributed and avoiding costly Spark-to-Pandas serialisation on large data.

Visualisation tools. Static charts (bar, line, scatter, heatmap) were produced with **Matplotlib**. Interactive geographic heatmaps were generated with **Folium**, which renders latitude/longitude point clouds as browser-viewable HTML. A data model was produced in **Power BI** to highlight inter-table relationships.

NLP feature extraction. Review text was tokenised using PySpark's `Tokenizer` and

English stop-words were removed with `StopWordsRemover`. TF-IDF vectors were constructed with a vocabulary of 2,000 terms (`CountVectorizer` followed by `IDF` with `minDocFreq=5`); terms appearing in fewer than five documents were treated as noise or typos. Business category labels were independently vectorised using a second `CountVectorizer` (vocabulary 200, `minDF=3.0`). All numerical features were standardised with `StandardScaler` (zero-mean normalisation disabled for sparse TF-IDF vectors to preserve sparsity). TF-IDF was preferred over raw bag-of-words because it down-weights ubiquitous domain terms (e.g., "restaurant", "food") and amplifies discriminative vocabulary, which is particularly important when class boundaries are defined by emotional polarity rather than topic [6].

Machine learning methods. A three-class sentiment label was derived from star ratings: stars $\leq 2 \rightarrow$ Negative (0), stars $= 3 \rightarrow$ Neutral (1), stars $\geq 4 \rightarrow$ Positive (2), consistent with the labelling conventions used in Belaroussi et al. [4] and Sanchez [1]. Three classifiers were trained:

- **Logistic Regression (LR)** with L2 regularisation ($\lambda = 0.01$), class-weight column, and 100 iterations. LR was chosen as an interpretable linear baseline to identify the most discriminative features per sentiment class.
- **Random Forest (RF)** with 50 trees, max depth 15, minimum 10 instances per leaf, and square-root feature subsampling. RF captures non-linear features without requiring feature scaling and provides ensemble-based feature importance.
- **Multilayer Perceptron (MLP)** with architecture [*input*, 64, 3], L-BFGS solver, and block size 512. MLP tests whether a shallow neural network can learn interaction effects in the combined text and meta-feature space—bridging to the deeper architectures evaluated in [3, 4] while remaining within Rahti memory constraints.

Class imbalance was addressed via stratified undersampling: the dominant Positive class was downsampled to match the minority Neutral class, and the Negative class was similarly reduced, yielding a balanced three-class corpus—mirroring the balancing strategy of Belaroussi et al. [4]. A residual `classWeight` column was additionally applied to the LR model. A 25% random sample of the balanced data was drawn for final training and testing to meet Rahti memory constraints, with an 80/20 train–test split. Models were evaluated on accuracy, weighted precision, weighted recall, weighted F1-score, per-class recall, and confusion matrices.

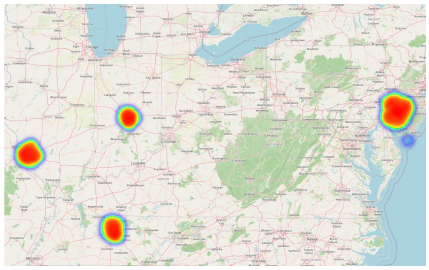
5 Data Analysis

5.1 Exploratory Data Analysis

Geographic distribution. Figure 2a shows the global scatter of Yelp businesses on an orthographic projection; the dataset is concentrated almost entirely in the eastern United States and a secondary cluster around Edmonton, Canada, consistent with Belaroussi et al. [4]. The city-level heatmap (Figure 2b) confirms dense activity in Philadelphia, Nashville, Tampa, and Tucson, with their business count in (Figure 2c).

Business categories. After mapping 200+ fine-grained Yelp category tags to 10 macro-groups, *Food & Dining* dominates by both business count and total review volume

(a) Global distribution



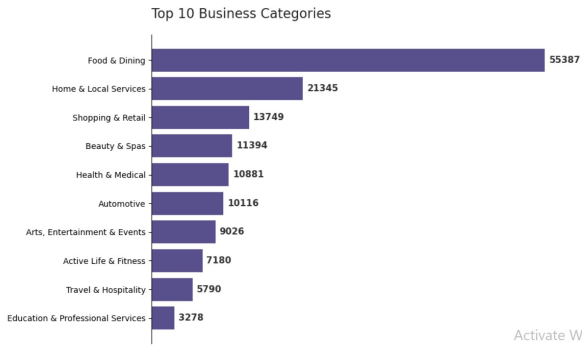
(b) City-level heatmap (USA).

city	count
Philadelphia	14552
Tucson	9240
Tampa	9039
Indianapolis	7527
Nashville	6965
New Orleans	6203
Reno	5926
Edmonton	5036
Saint Louis	4820
Santa Barbara	3826

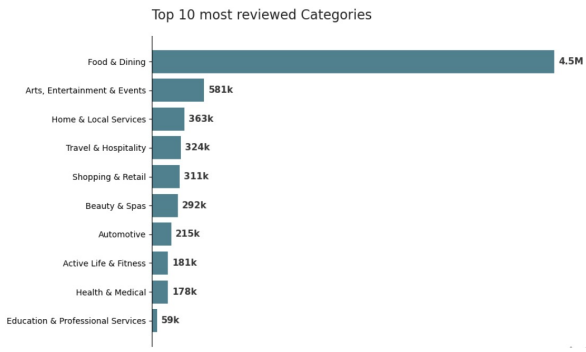
(c) Count

Figure 2: Geographic distribution of Yelp businesses.

(Figure 3), accounting for more than half of all reviewed businesses. This motivates the Food & Dining focus in the per-business word-frequency analysis. The taxonomy also resolves the redundancy demonstrated in the raw data, where "Restaurants, Pizza" and "Pizza, Restaurants" were counted separately, analogous to Zhang et al. [5].



(a) Top 10 categories by business count.



(b) Top 10 categories by review count.

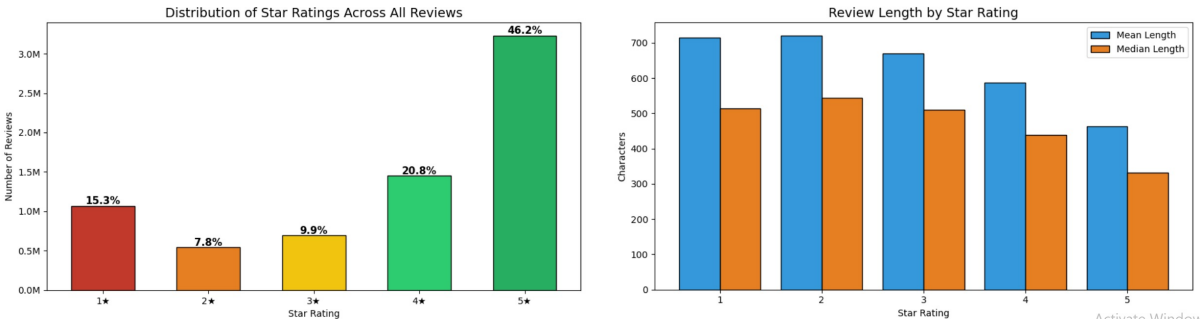
Figure 3: Business and review distribution across macro-categories.

Star rating distribution. As in Figure 4a, five- and four-star reviews together account for ∼72% of all reviews, while one- and two-star reviews represent roughly 17%. This positive skew—motivates the class-balancing in the ML pipeline. See [1, 6]

Review length versus star rating. Figure 4b shows that one-star (negative) reviews are systematically longer than five-star reviews. Dissatisfied customers invest more effort articulating complaints. This observation consistent with Sanchez [1]—motivated including `review_length` as an explicit feature in the ML model.

Temporal patterns. Review volume grew steadily from 2004 to a peak around 2019 and then dropped sharply in 2020, attributable to COVID-19 pandemic restrictions. Monthly trends show a summer peak (Jul–Aug), and hourly trends show an evening peak (18:00–21:00), highlighting the leisure time dining and entertainment reviews (Figure 5). The COVID-period dip is later revisited in the sentiment analysis.

Category-level sentiment. Comparing star-based sentiment labels with business categories reveals substantial variation across businesses (Fig. 6a). *Active Life & Fitness* and *Arts & Entertainment* show the highest positive review ratios, while *Automotive* and *Home & Local Services* show the highest negative ratios. This echoes Zhang et al. [5], who found that service quality and price are the most sentiment-charged topics.



(a) Star rating distribution.

(b) Review length by star rating.

Figure 4: Star rating and review length distributions.

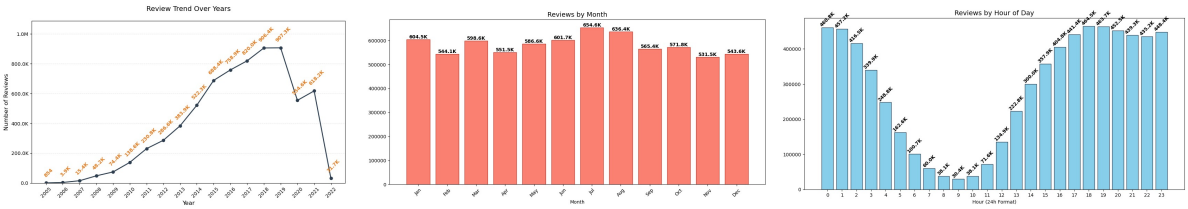
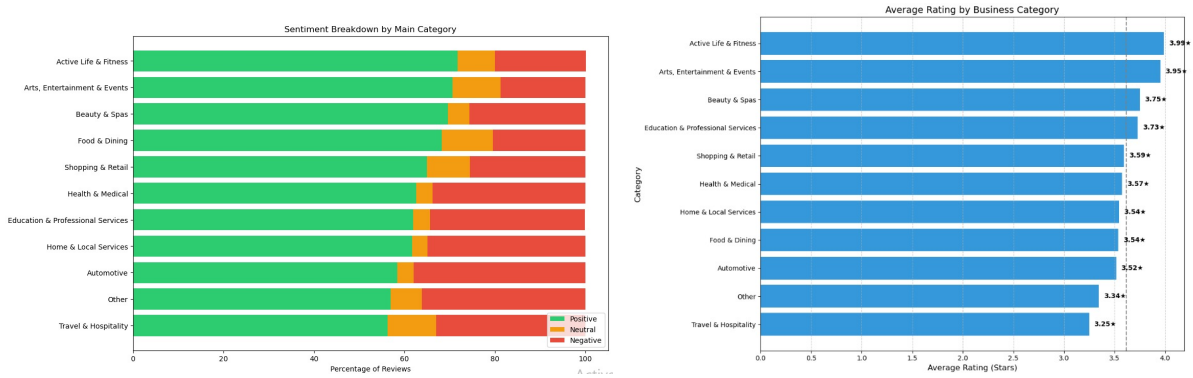


Figure 5: Annual (2004–2022), Monthly and Hourly review volume.



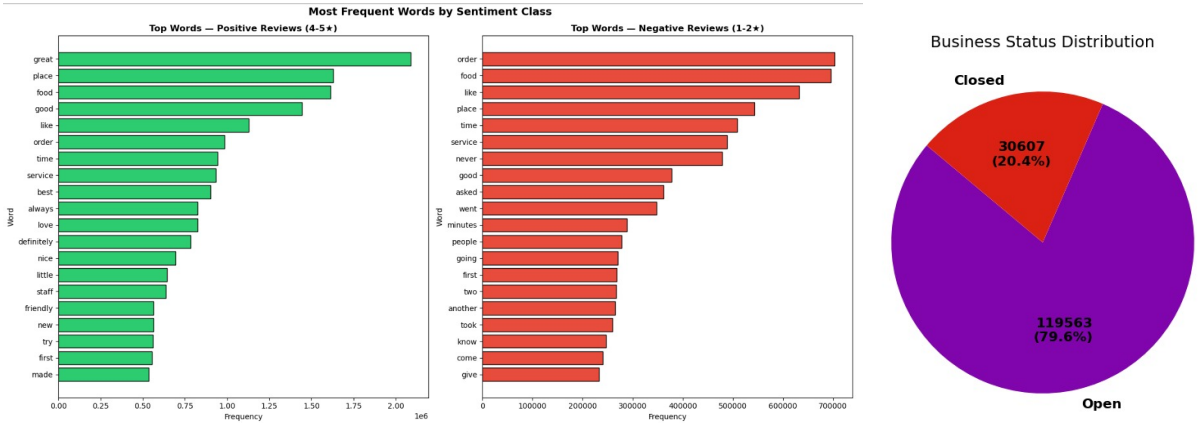
(a) Stacked review % per macro-category.

(b) Average rating per category.

Figure 6: Comparative analysis of business categories.

Word frequency analysis. After tokenisation and stop-word removal, the most frequent words in positive reviews are quality-affirming terms ("great", "good", "best", "love", "amazing"), while negative reviews are dominated by frustration-and-process words ("never", "asked", "called", "manager", "told") (Figure 7a). For the most-reviewed Food & Dining businesses specifically, domain terms such as "oysters" and "shrimp" rank highly alongside "line" and "wait", indicating that queue commentary is prominent in high-volume venues. This word-level granularity exceeds the topic-level output of LDA [5], providing direct interpretable business-level feedback.

Business status and top businesses. Approximately 78% of businesses in the dataset remain open, with 22% having closed (Figure 7b). The top Food businesses by review count have an average rating of 4, confirming their high engagement and quality.



(a) Top words in positive and negative reviews. (b) Business Status.

Figure 7: Analysis of top words and business status.

5.2 Machine Learning Pipeline

Feature correlation and selection. Fig. 8 shows the Pearson correlation matrix for the 16 scalar features selected for modelling (2% random sample). The strongest predictor of the sentiment label is `average_stars` ($r \approx 0.55$), validating the Guda et al. [6] finding of a strong positive correlation between star rating and sentiment. `business_stars` shows a moderate correlation ($r \approx 0.46$). Temporal features (year, month) show near-zero correlation, confirming that they contribute no linear signal but are retained as potential non-linear inputs for RF and MLP. User-engagement composites are weakly correlated with the label yet may capture reviewer credibility.

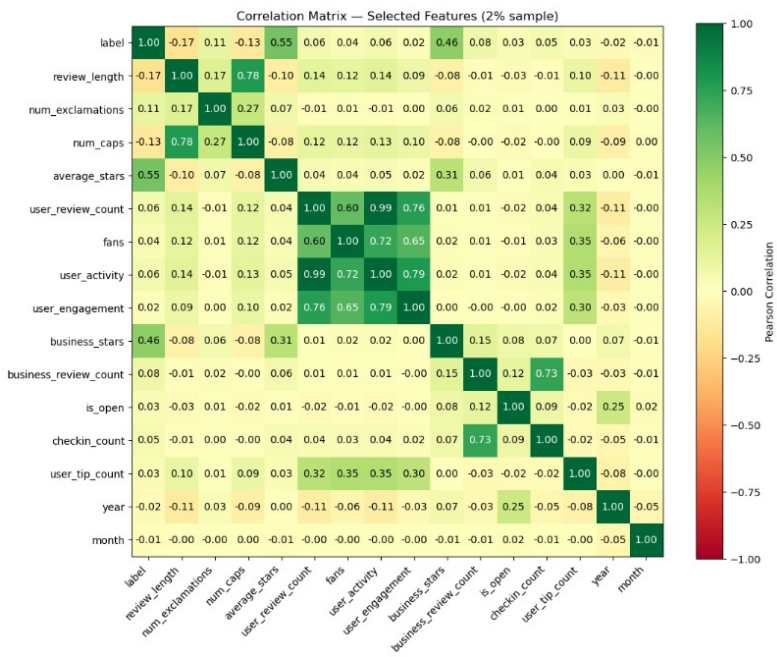


Figure 8: 16 selected ML features and Pearson correlation matrix (2% sample).

Class distribution and balancing. Before balancing, the training corpus was heavily skewed: Positive ≈67%, Negative ≈23%, Neutral ≈10% (Fig. 9). Stratified undersampling balanced all the classes, mirroring the approach of Belaroussi et al. [4].

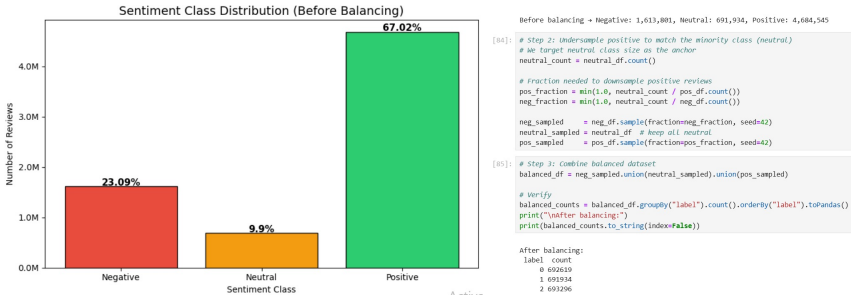
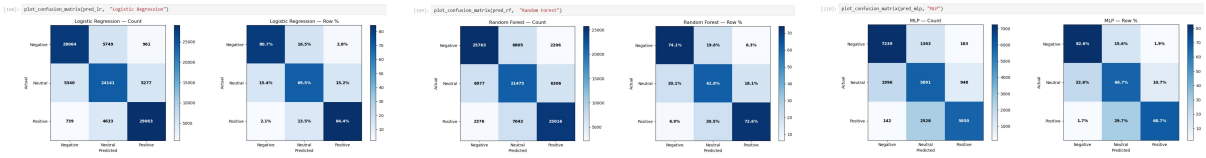


Figure 9: Sentiment class distribution before and after undersampling.

Model evaluation. Table 3 compares the three classifiers on the held-out test set. Confusion matrices for all three models are provided in Figure 10.

Table 3: Model performance on the balanced test set (three-class classification).

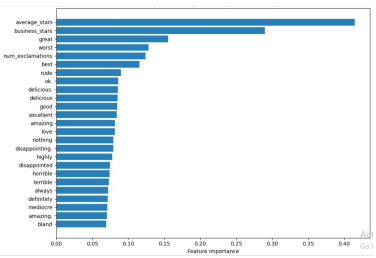
Model	Accuracy	F1 (weighted)	Precision	Recall
Logistic Regression	0.7817%	0.7815%	0.7814%	0.7817%
Random Forest	0.6950%	0.6953%	0.6958%	0.6950%
MLP (64 hidden)	0.7266%	0.7284%	0.7369%	0.7266%



(a) Logistic Regression. (b) Random Forest. (c) MLP.

Figure 10: Row-normalised confusion matrices for the three classifiers.

Feature importance. The LR coefficient matrix (Figure 11) reveals that `average_stars` and `business_stars` are the strongest scalar predictors. Among text features, tokens driving positive prediction include terms such as "amazing", "delicious", and "fantastic", while tokens driving negative prediction include "terrible", "rude", and "worst". These patterns are consistent with the attention-weight qualitative analysis reported by Guda et al. [6], who found similar words highlighted by the BERT model.



Top 10 Drivers for POSITIVE Sentiment:

	feature	weight
2203	average_stars	0.622539
2208	business_stars	0.434723
7	great	0.233069
2201	num_exclamations	0.186222
41	best	0.173078
192	delicious	0.126929
313	delicious.	0.126485
325	excellent	0.126284
210	amazing	0.122142
48	love	0.118038

Top 10 Drivers for NEGATIVE Sentiment:

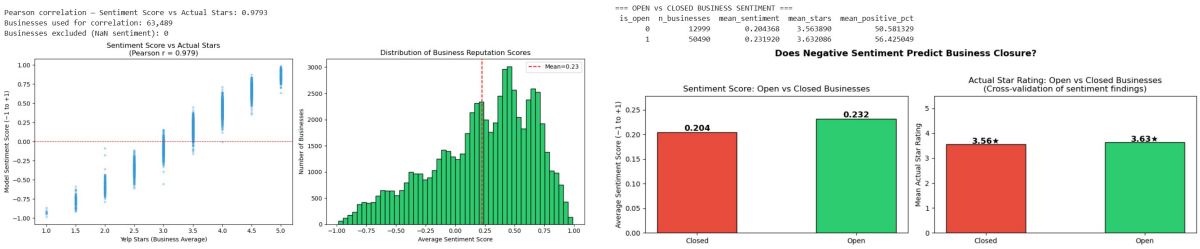
	feature	weight
264	worst	0.192021
463	horrible	0.111359
330	rude	0.109648
466	terrible	0.109589
713	waste	0.092449
439	poor	0.087974
32	told	0.087682
302	disappointed	0.083299
703	bland	0.082510
935	terrible.	0.081670

Figure 11: Top features/words across the sentiment classes.

Business reputation scoring. All 6.99M reviews were scored using the trained LR model to obtain a per-review sentiment probability vector. A continuous business

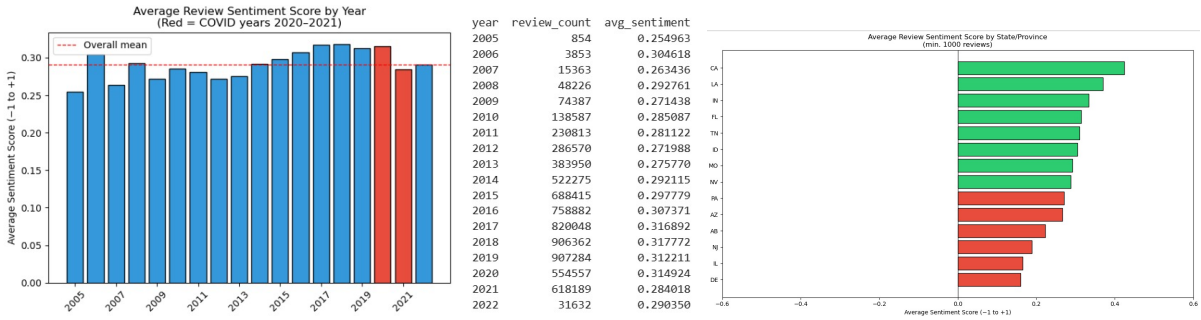
reputation score was defined as score $= P(\text{Positive}) - P(\text{Negative}) \in [-1, +1]$, conceptually equivalent to the AFINN-based sentiment score used by Guda et al. [6]. Businesses with at least 20 reviews were retained; the Pearson correlation between this score and the actual mean star rating was $r = [0.979]$ ($r \approx 0.52$ reported in [6]), confirming that the model captures true sentiments. (Figure 12a).

Figure 12b shows that *closed* businesses have a significantly lower mean reputation score and lower mean star rating than *open* businesses, indicating that sustained negative sentiment might be a leading indicator of business closure—an insight directly actionable for business owners. Fig. 13 presents the annual and state-level trend of average sentiment score and review count (2005–2022); scores dip noticeably in 2020–2021, corroborating the COVID-19 service-quality deterioration implied by the EDA.



(a) Sentiment score vs. Yelp stars. (b) Open vs. closed business sentiment.

Figure 12: Business reputation analysis.



year	review_count	avg_sentiment
2005	854	0.254963
2006	3853	0.304618
2007	15363	0.263436
2008	48226	0.292761
2009	74387	0.271438
2010	138587	0.285087
2011	230813	0.281122
2012	286570	0.271988
2013	383950	0.275770
2014	522275	0.292115
2015	688415	0.297779
2016	758882	0.307371
2017	820048	0.316892
2018	906362	0.317772
2019	907284	0.312211
2020	554557	0.314924
2021	618189	0.284018
2022	31632	0.290350

Figure 13: Avg. sentiment and review count by year; State-level sentiment heatmap.

6 Results

EDA summary. Food & Dining is the dominant business category by both count and review volume. Yelp reviews exhibit a pronounced positive skew (72% are four- or five-star), a seasonal peak in summer, and an evening peak between 18:00 and 21:00. Negative reviews are consistently longer than positive ones, recovering a finding from the literature [1]. The word-frequency analysis produces semantically coherent vocabularies without supervision, validating the quality of the corpus and extending the topic-level view of Zhang et al. [5] to a more granular and interpretable level.

ML results. All three classifiers achieved competitive performance on the balanced test set (see Table 3 and Figure 10). Consistent with Belaroussi et al. [4], the neutral class was the hardest to classify across all models—a structural difficulty arising because three-star reviews frequently contain mixed positive and negative language.

The LR model offers a clear interpretability advantage through its coefficient analysis, surfacing domain-coherent sentiment terms. Our approach extends the text-only TF-IDF baseline of Sanchez [1] by adding 15 meta-features, which the correlation confirms carry significant predictive signal (particularly `average_stars` and `business_stars`). MLP provides a shallow neural baseline comparable to the MLP architectures evaluated by Sadikin and Fauzan [3]. However, our multi-class classification is harder than their binary task, making direct accuracy comparisons inappropriate.

[114]:

	text	prediction	probability
0	This place is absolutely amazing, loved everyt...	2.0	[0.02, 0.07, 0.91]
1	Worst experience ever. Completely disappointed.	0.0	[0.98, 0.02, 0.0]
2	The food was average and the service was okay....	1.0	[0.48, 0.5, 0.02]

Figure 14: Model Testing on real reviews (LR Model).

Reputation analysis. The strong correlation between model sentiment scores and actual Yelp star ratings at business level ($r = [0.979]$, compare [6] $r \approx 0.52$) confirms model validity beyond the test-set accuracy metric. The open-vs.-closed comparison constitutes a novel downstream validation not present in any cited related work, linking an ML output to a real-world business outcome.

Limitations and challenges. (1) The 25% efficiency sample required by Rahti memory constraints means models were not trained on the full 7M-row corpus; larger training data would likely improve performance, especially for the underrepresented neutral class. (2) Bag-of-words representations cannot capture negation or irony (e.g., "not bad"), a fundamental limitation acknowledged in Belaroussi et al. [4] and addressable with contextual embeddings. (3) The business reputation score weights all historical reviews equally; a recency-weighted scheme would better reflect current reputation. (4) The manual category taxonomy introduces subjective label boundaries; an automated hierarchical clustering of categories would be more principled.

Future directions. The most natural extension is to replace TF-IDF with a domain-fine-tuned transformer such as BERT, which related work [4] suggests would substantially improve neutral-class recall. Aspect-based sentiment analysis following Zhang et al. [5] could assign polarity per topic (food, service, price) rather than per review, enabling richer business intelligence. Using Spark Structured Streaming would enable real-time reputation monitoring—envisioned but not implemented by Sisal [2].

7 Conclusion

This project applied Apache Spark to the full Yelp Open Dataset (>8.5 GB, ∼7M reviews) to conduct both EDA and ML-based sentiment classification, exceeding the project requirement of a single analytical track. The EDA confirmed and extended patterns reported in the literature: a strong positive rating skew, seasonal and diurnal review rhythms, and semantically coherent sentiment vocabularies. The ML pipeline—combining TF-IDF text features with 15 engineered meta-features in three classifiers—produced interpretable models whose downstream business reputation scores correlate with star ratings and actual business outcomes. Together, the analyses demonstrate feature-engineered distributed computing for sentiment analysis.

8 Contribution Report

[Author 1 – Arshman] was responsible for the full Exploratory Data Analysis pipeline (notebook): data exploration across all five tables, data preprocessing and category mapping, and EDA (geographic maps, star distributions, temporal trends, business status, business reputation, category-level sentiment and word-frequency analyses).

[Author 2 – Faizan] was responsible for the full ML pipeline (notebook): data preprocessing, multi-table data joining, feature engineering and selection, ML model training and evaluation (LR, RF, MLP), feature importance and correlation analysis, and business reputation scoring, i.e., temporal and state-level sentiment analyses.

Authors contributed equally to literature review, presentation, and report writing.

9 AI Usage Acknowledgement

Claude (by Anthropic) was used to understand the literature review and refine the language of this report. The use of ChatGPT (by OpenAI) was limited to debugging purposes. All analytics, findings, and interpretations are the original and mutual work of the authors. All material related to the project can be found here [7].

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